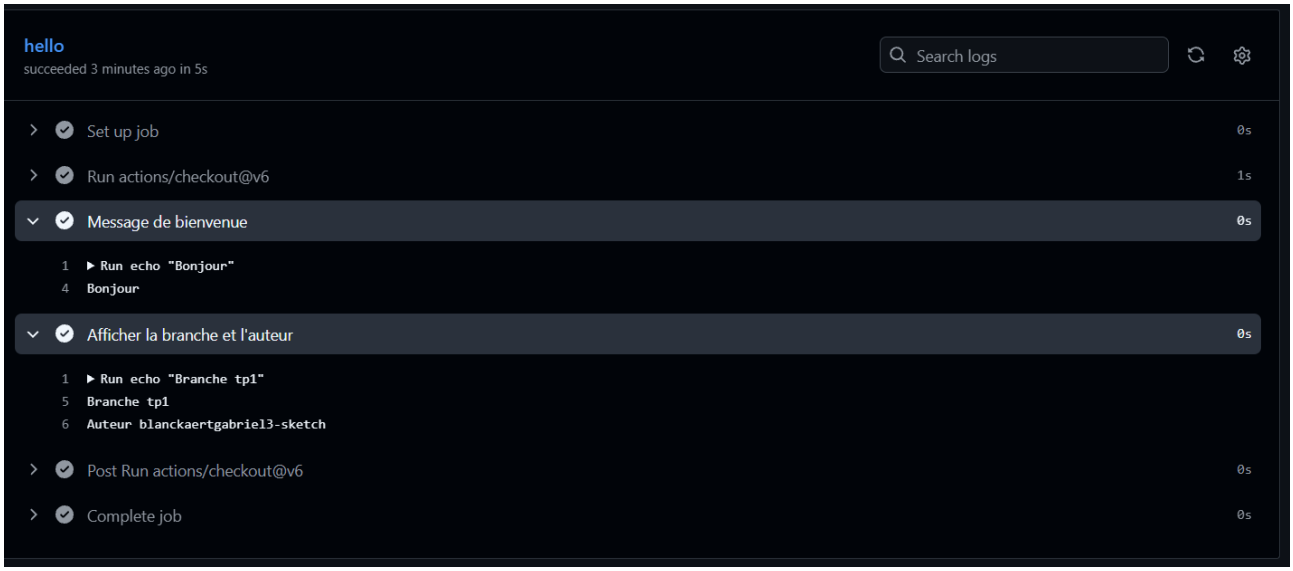
Gabriel Blanckaert
02/11/2026
https://github.com/blanckaertgabriel3-sketch/tasks-api

TP1

TP1-1



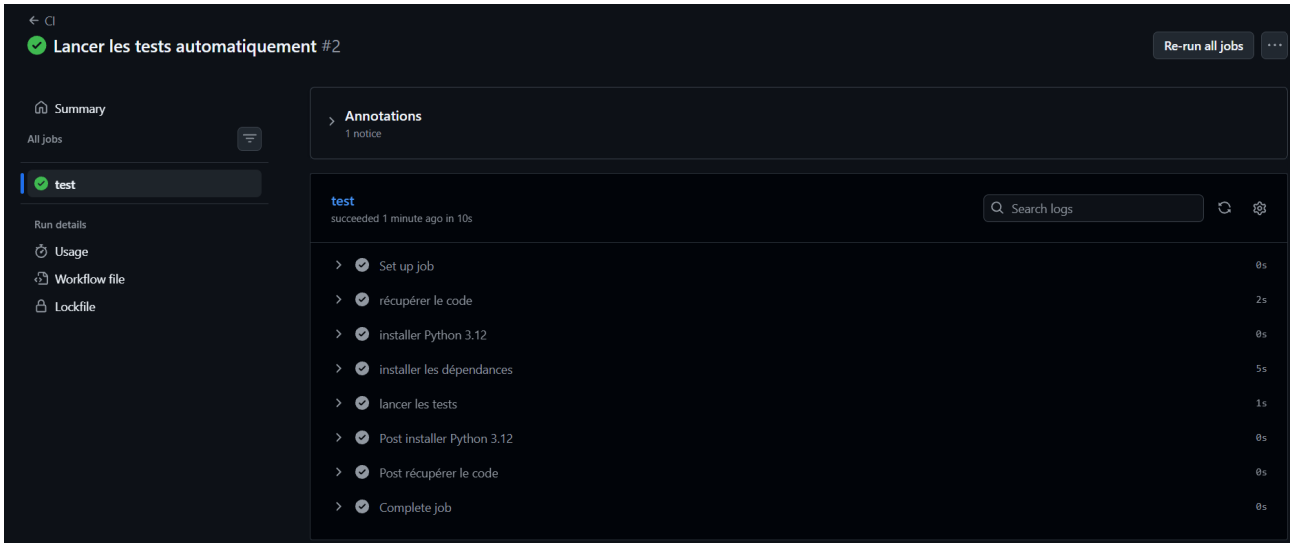
Combien de temps a duré le run « Hello » ?

5 secondes pour le run hello

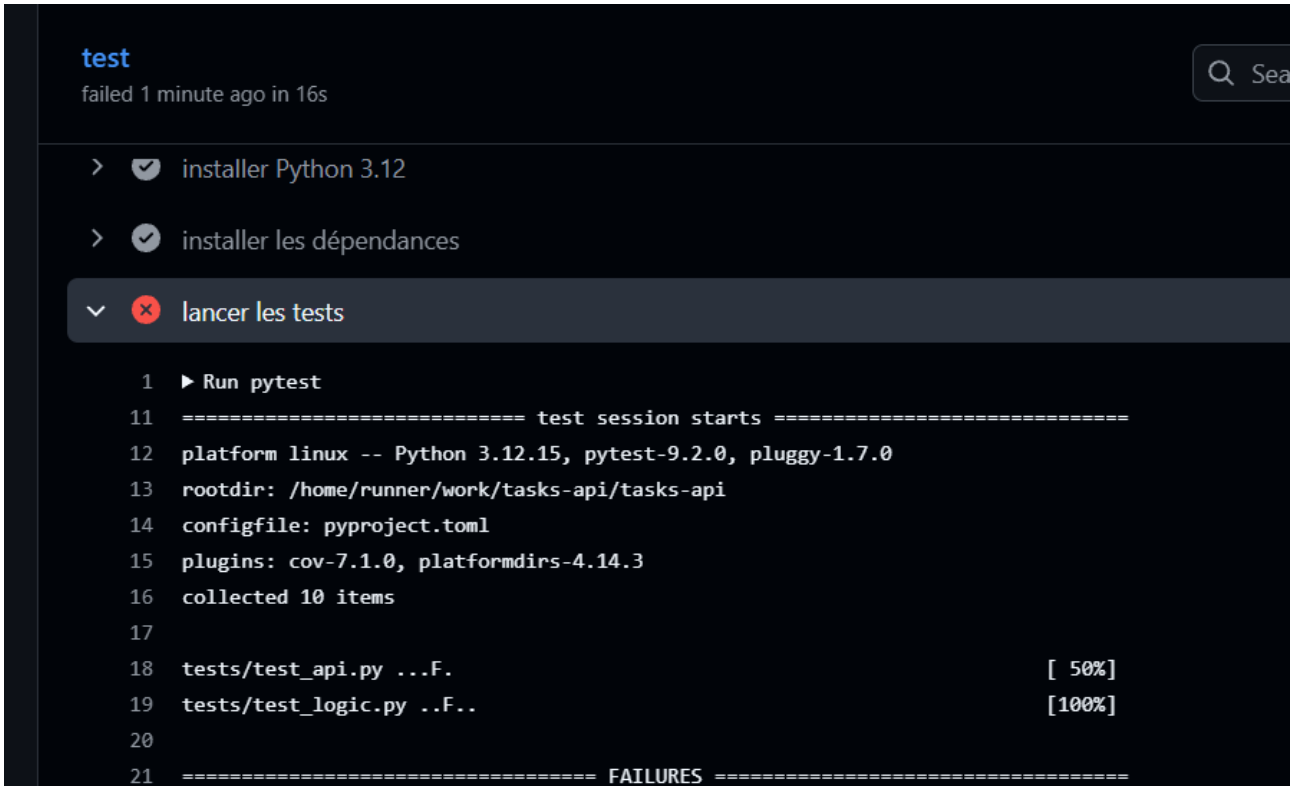
Où voit-on les logs de chaque étape ?

dans action, le run hello, le job hello

TP1-2



TP1-3



Quel test a échoué dans la partie 3, et à quelle ligne ?

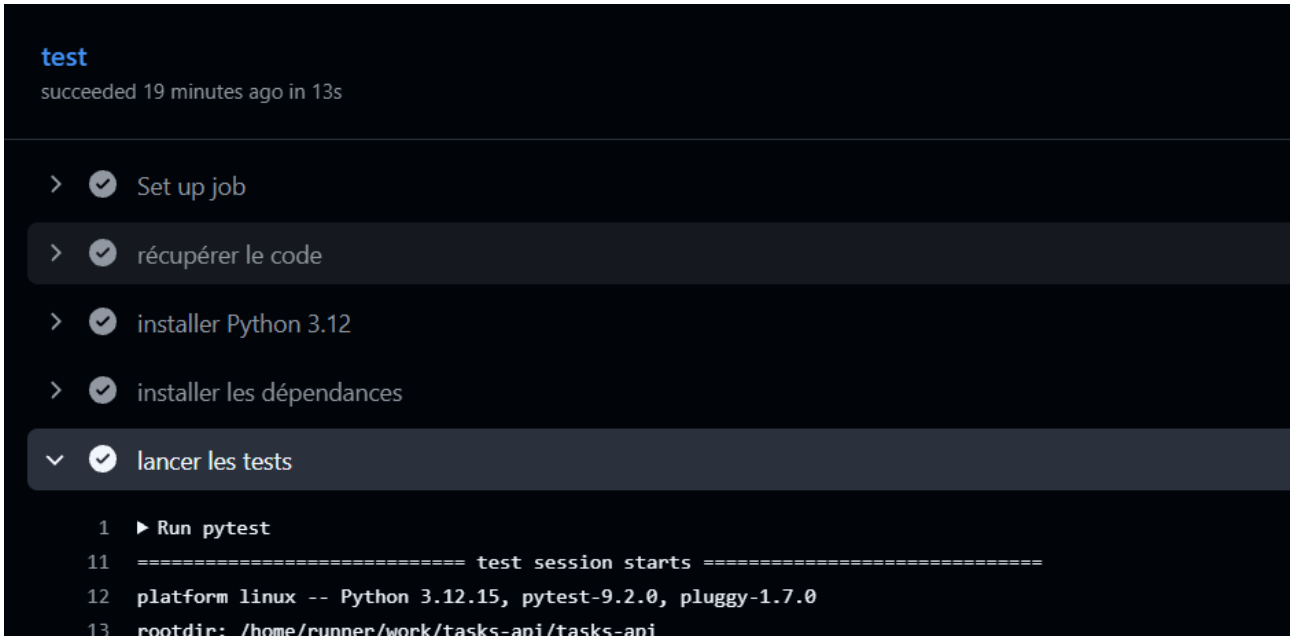
tests/test_api.py : ligne 29, test test_add_and_list_tasks.

tests/test_logic.py : ligne 13, test test_is_valid_title_accepts_normal_title

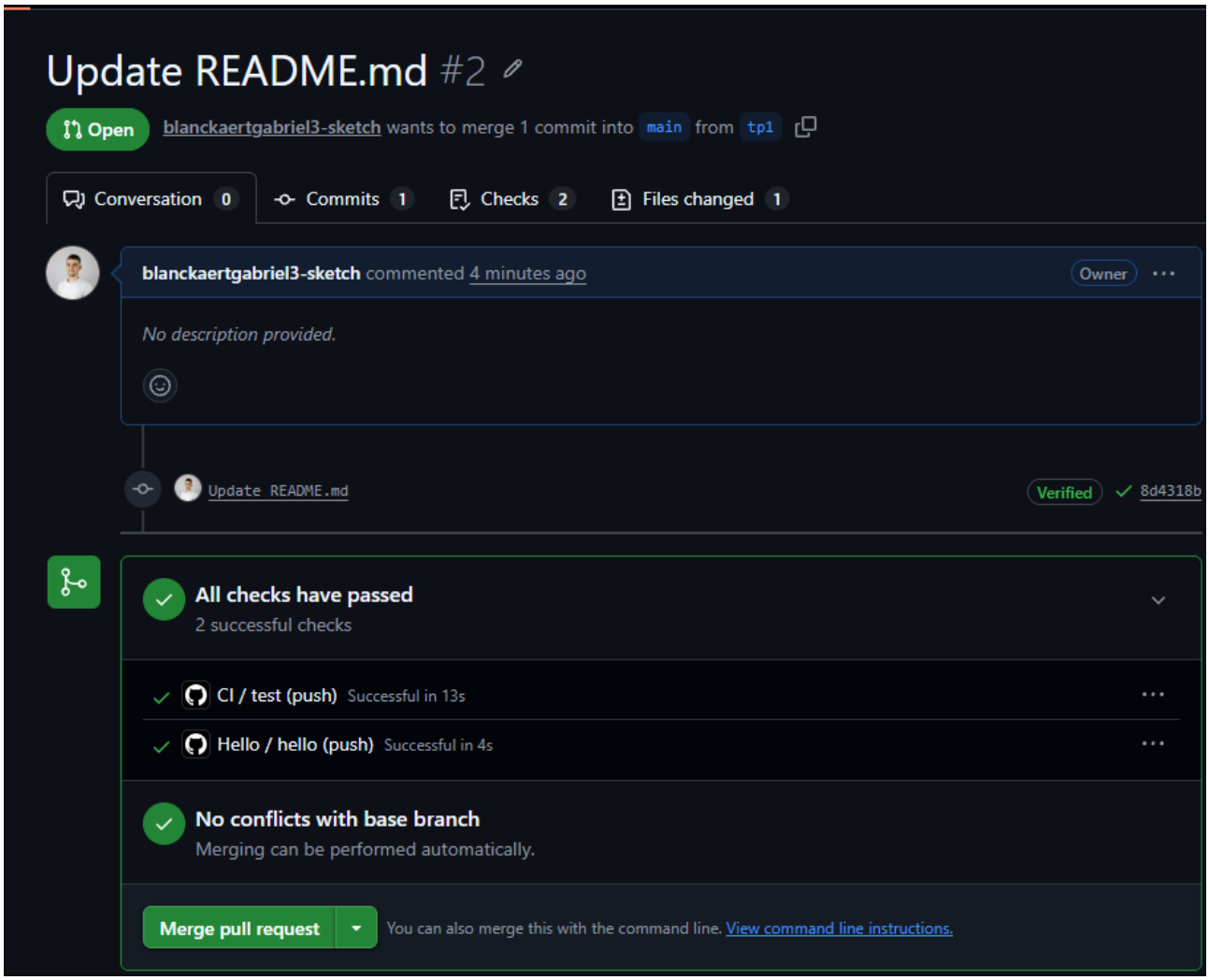
Qu'est-ce qui a permis de le trouver rapidement ?

grâce au rapport de pytest, qui indique le nom du test, le fichier et le numéro de ligne

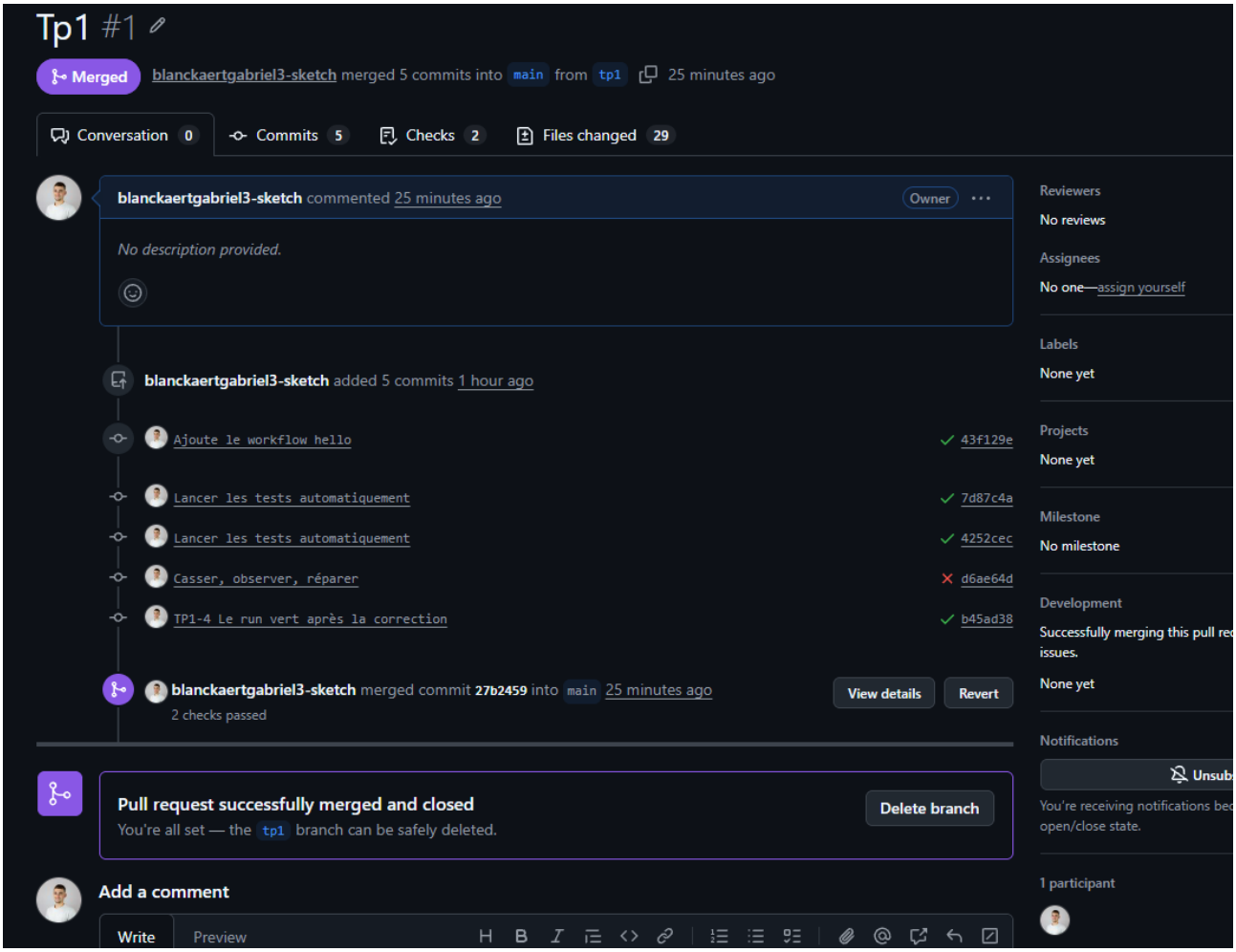
TP1-4



TP1-5


Update README.md #2
Open
blanckaertgabriel3-sketch wants to merge 1 commit into main from tp1
Conversation 0
Commits 1
Checks 2
Files changed 1
blanckaertgabriel3-sketch commented 4 minutes ago
Owner
No description provided.
Update README.md
Verified
8d4318b
All checks have passed
2 successful checks
CI / test (push) Successful in 13s
Hello / hello (push) Successful in 4s
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.

TP1-6


Tp1 #1
Merged
blanckaertgabriel3-sketch merged 5 commits into main from tp1 25 minutes ago
Conversation 0
Commits 5
Checks 2
Files changed 29
blanckaertgabriel3-sketch commented 25 minutes ago
Owner
No description provided.
blanckaertgabriel3-sketch added 5 commits 1 hour ago
Ajoute le workflow hello
43f129e
Lancer les tests automatiquement
7d87c4a
Lancer les tests automatiquement
4252cec
Casser, observer, réparer
d6ae64d
TP1-4 Le run vert après la correction
b45ad38
blanckaertgabriel3-sketch merged commit 27b2459 into main 25 minutes ago
2 checks passed
View details
Revert
Pull request successfully merged and closed
You're all set — the tp1 branch can be safely deleted.
Delete branch
Add a comment
Write
Preview
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull
issues.
None yet
Notifications
Unsub
You're receiving notifications
open/close state.
1 participant

TP1-7

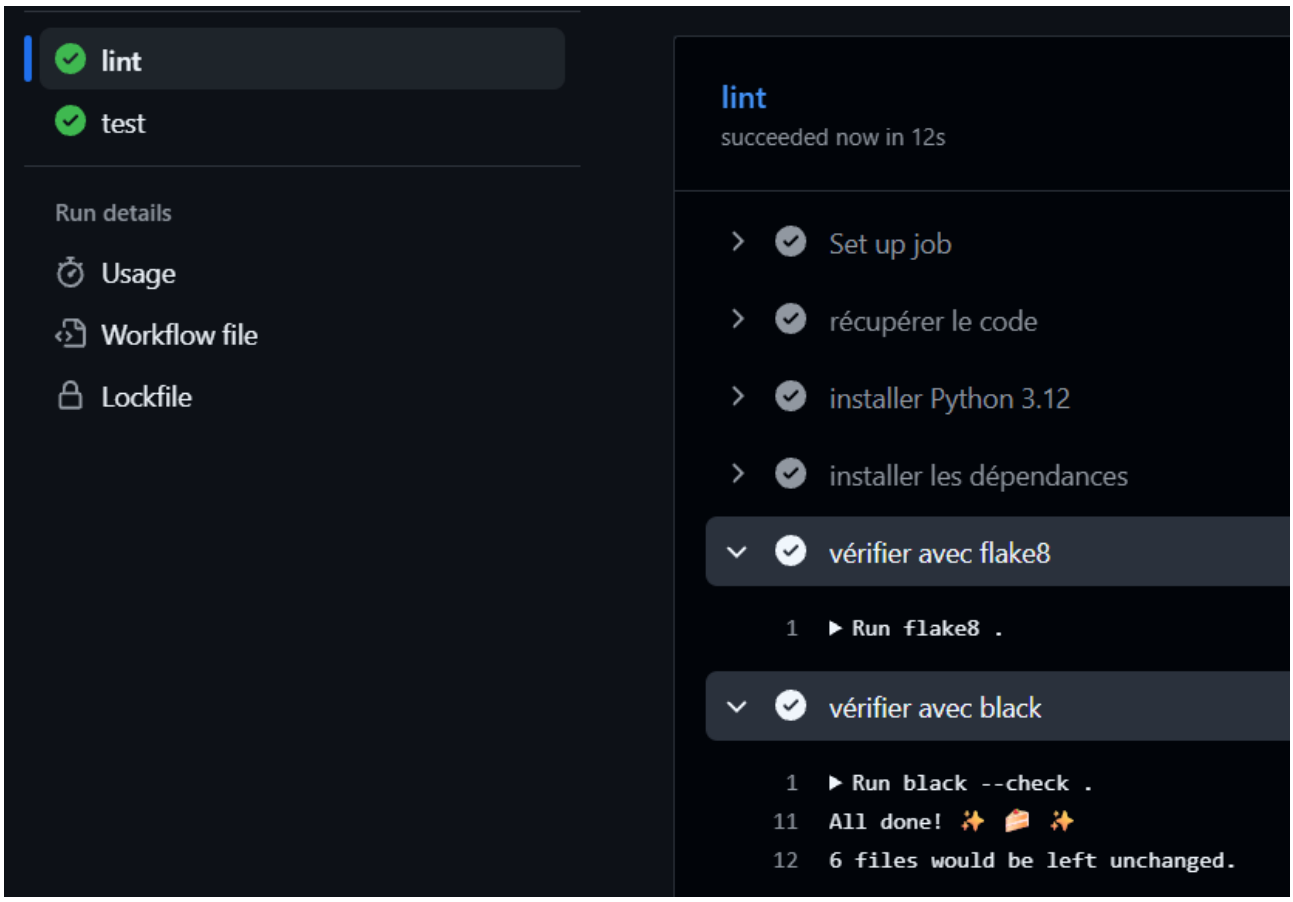


Pourquoi attend-on que la CI soit verte avant de fusionner ?

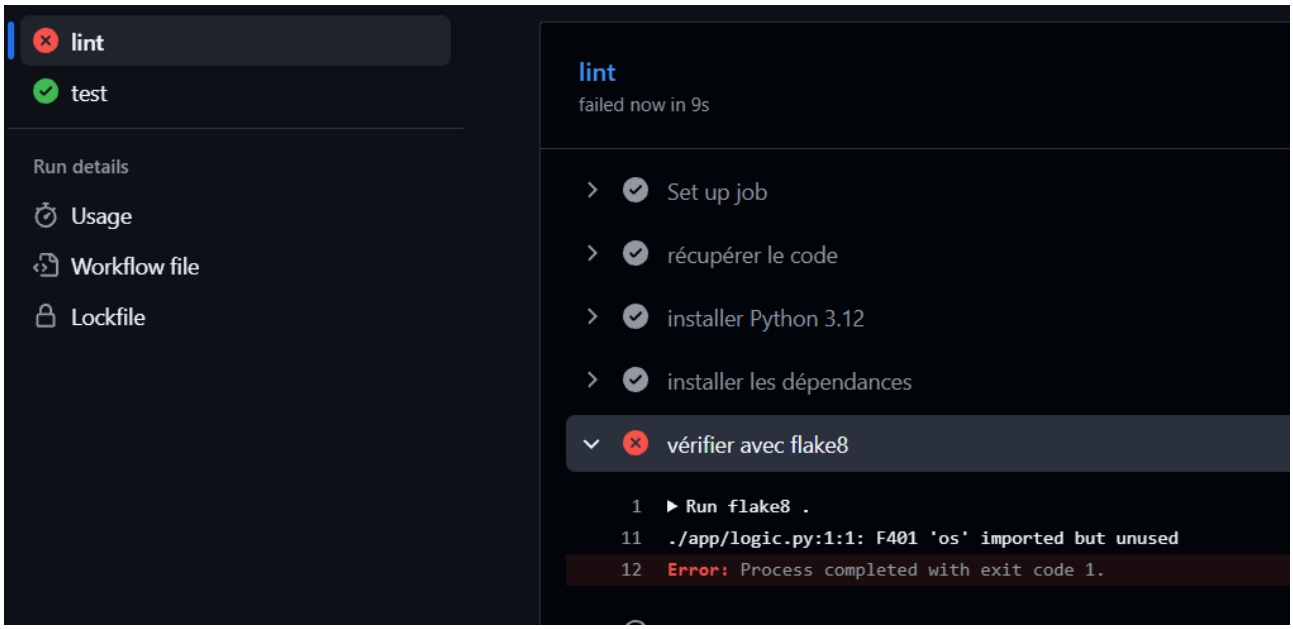
pour vérifier automatiquement que le code fonctionne et que tous les tests passent, pour éviter d'intégrer des erreurs dans la branche `main`.

TP2

TP2-1


lint
test
Run details
Usage
Workflow file
Lockfile
lint
succeeded now in 12s
Set up job
récupérer le code
installer Python 3.12
installer les dépendances
vérifier avec flake8
1 ▶ Run flake8 .
vérifier avec black
1 ▶ Run black --check .
11 All done! ✨ 🍰 ✨
12 6 files would be left unchanged.

TP2-2


lint
test
Run details
Usage
Workflow file
Lockfile
lint
failed now in 9s
Set up job
récupérer le code
installer Python 3.12
installer les dépendances
vérifier avec flake8
1 ▶ Run flake8 .
11 ./app/logic.py:1:1: F401 'os' imported but unused
12 Error: Process completed with exit code 1.

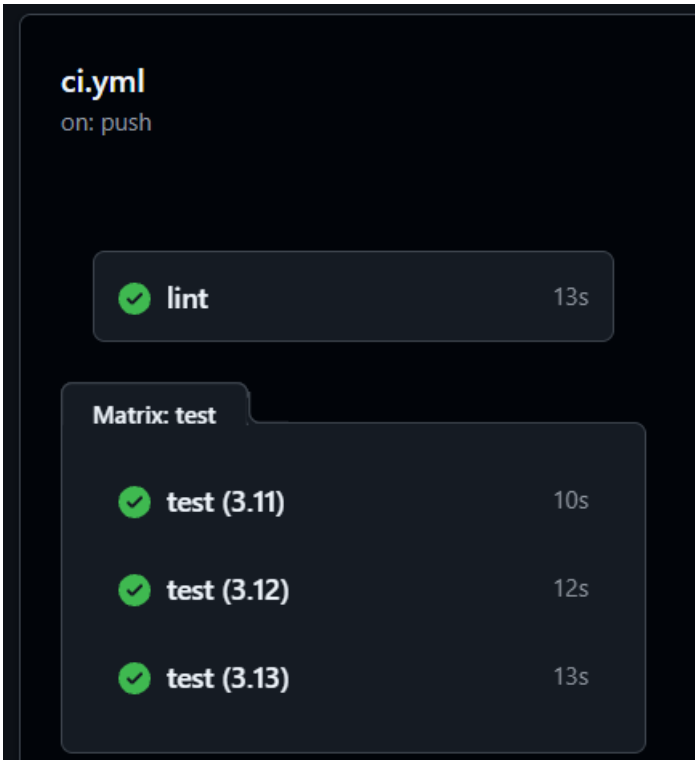
Quelle différence entre un linter (flake8) et un formateur (black) ?

flake détecte les problémes dans le code et black corrige automatiquement la mise en forme

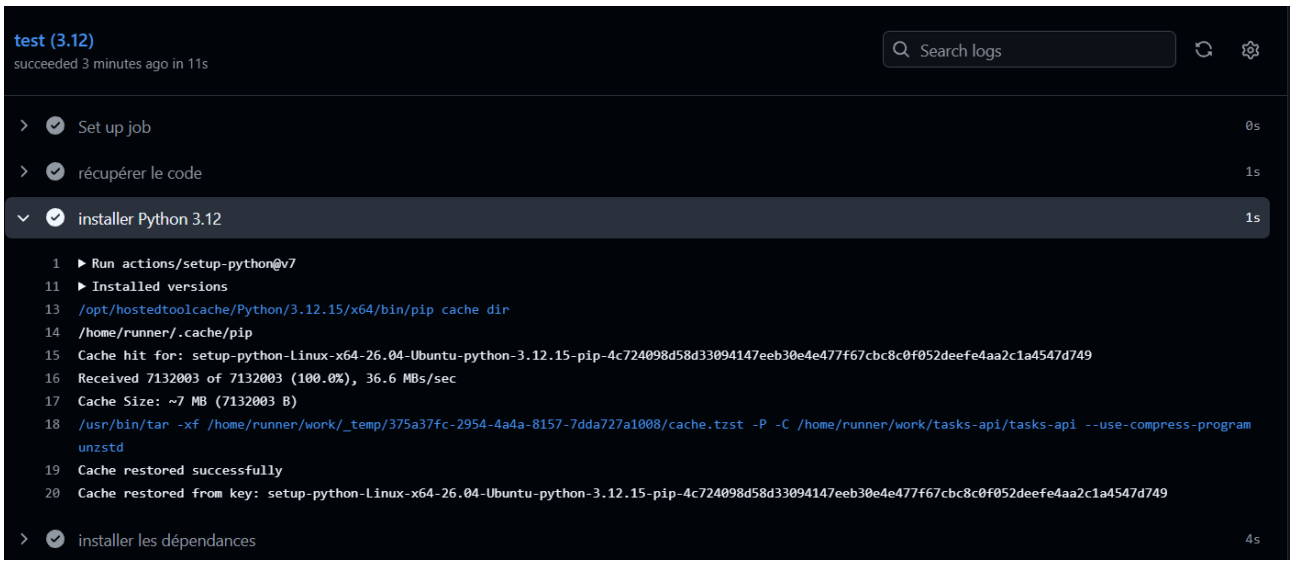
Pourquoi séparer `lint` et `test` en deux jobs ?

pour distinguer les erreurs de qualité de code des erreurs de fonctionnement

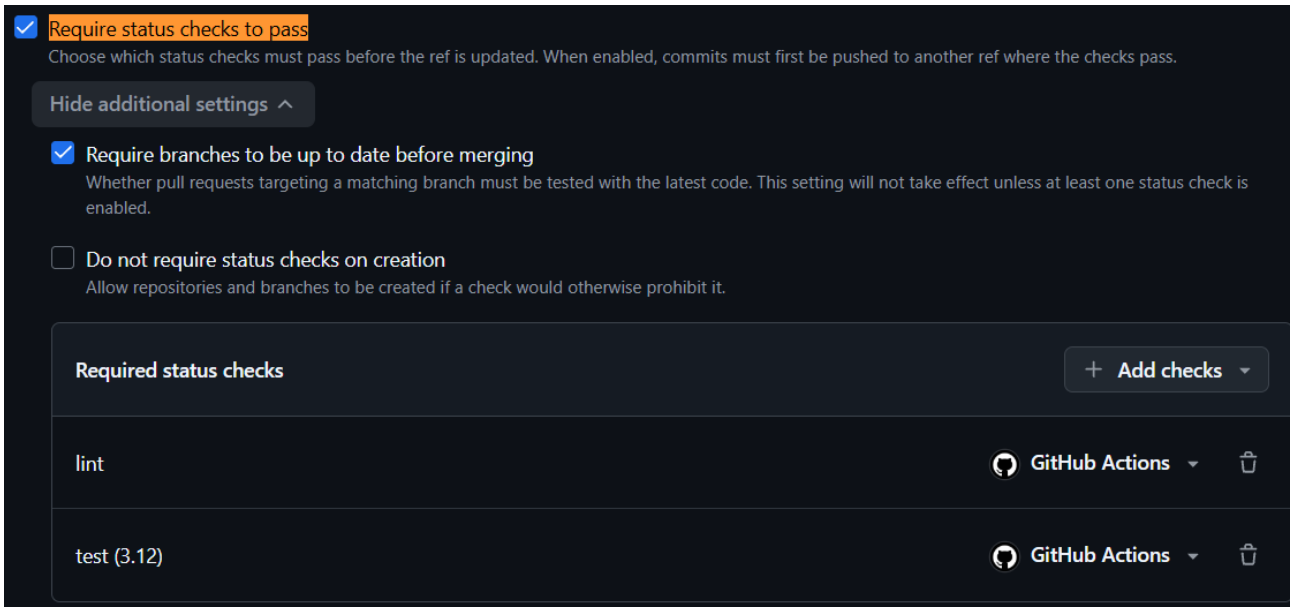
TP2-3



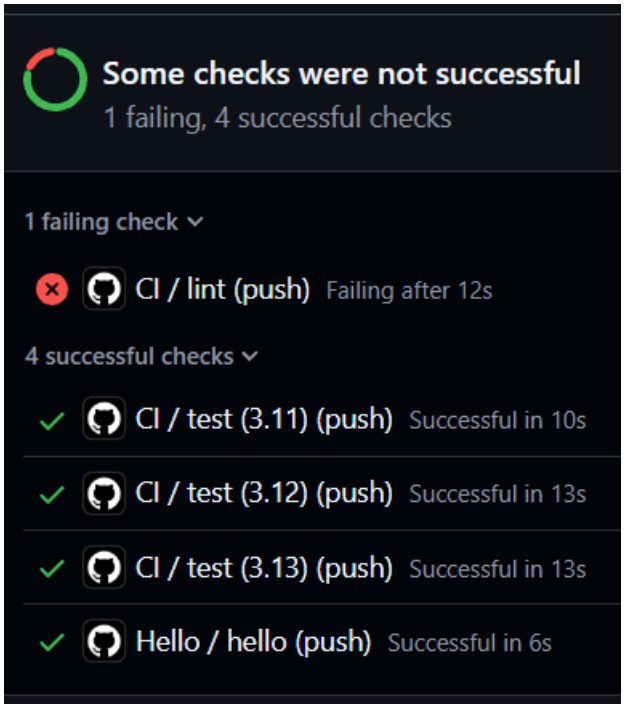
TP2-4

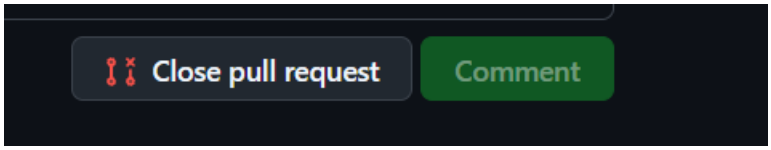

test (3.12)
succeeded 3 minutes ago in 11s
Search logs
Set up job
0s
récupérer le code
1s
installer Python 3.12
1s
1 ▶ Run actions/setup-python@v7
11 ▶ Installed versions
13 /opt/hostedtoolcache/Python/3.12.15/x64/bin/pip cache dir
14 /home/runner/.cache/pip
15 Cache hit for: setup-python-Linux-x64-26.04-Ubuntu-python-3.12.15-pip-4c724098d58d33094147eeb30e4e477f67cbc8c0f052deefe4aa2c1a4547d749
16 Received 7132003 of 7132003 (100.0%), 36.6 MBs/sec
17 Cache Size: ~7 MB (7132003 B)
18 /usr/bin/tar -xf /home/runner/work/_temp/375a37fc-2954-4a4a-8157-7dda727a1008/cache.tzst -P -C /home/runner/work/tasks-api/tasks-api --use-compress-program unzstd
19 Cache restored successfully
20 Cache restored from key: setup-python-Linux-x64-26.04-Ubuntu-python-3.12.15-pip-4c724098d58d33094147eeb30e4e477f67cbc8c0f052deefe4aa2c1a4547d749
installer les dépendances
4s

TP2-5


Require status checks to pass
Choose which status checks must pass before the ref is updated. When enabled, commits must first be pushed to another ref where the checks pass.
Hide additional settings
Require branches to be up to date before merging
Whether pull requests targeting a matching branch must be tested with the latest code. This setting will not take effect unless at least one status check is enabled.
Do not require status checks on creation
Allow repositories and branches to be created if a check would otherwise prohibit it.
Required status checks
Add checks
lint
GitHub Actions
test (3.12)
GitHub Actions

TP2-6

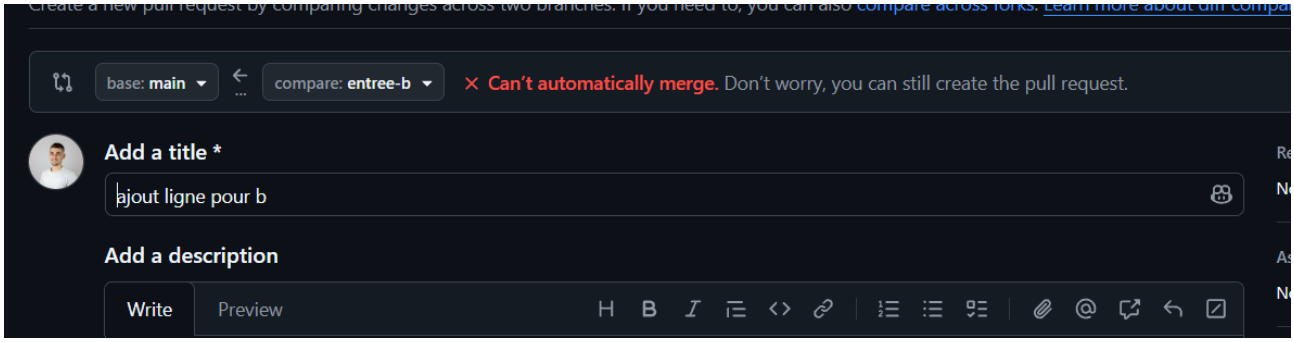


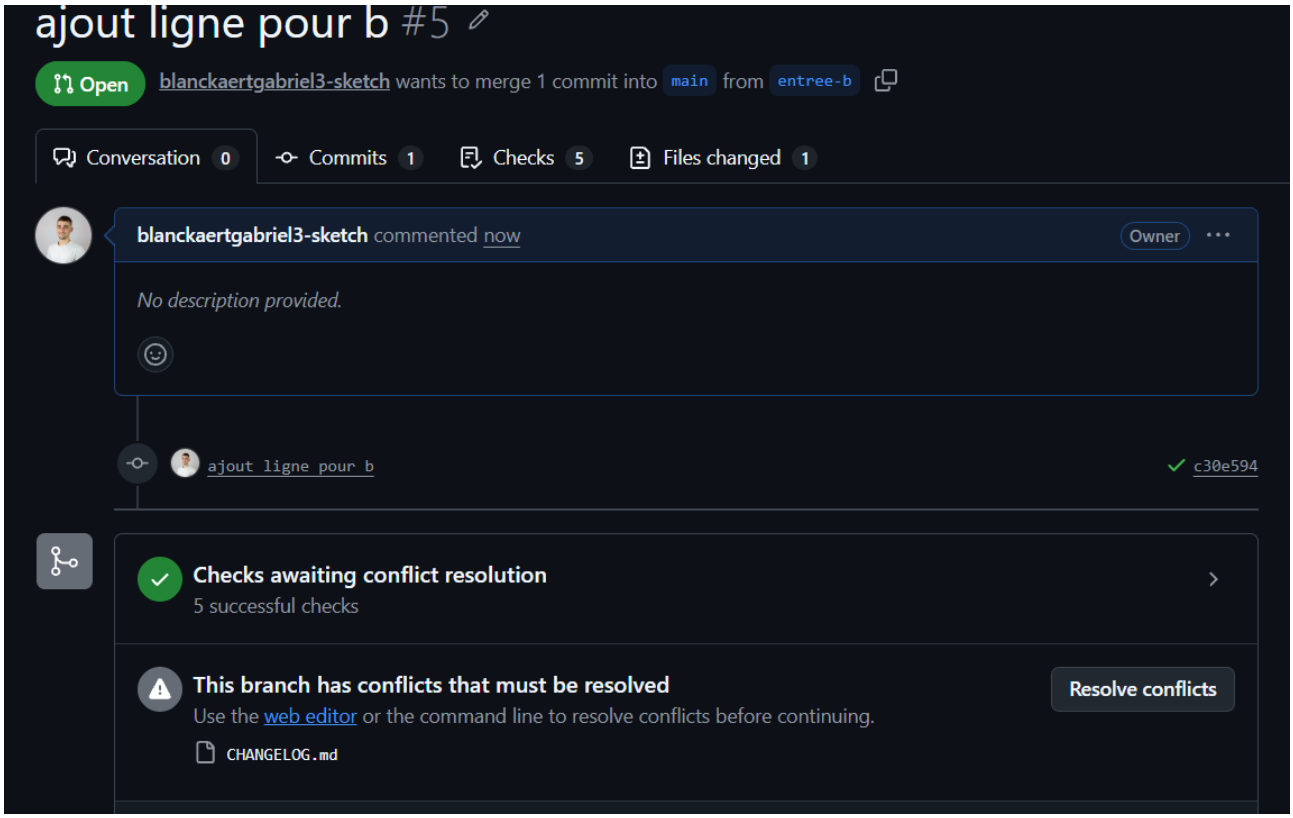


Pourquoi interdire de pousser directement sur `main` ?

Dans les paramètres de la branche main, j'ai imposer la réussite des testes CI avant de push sur main.

TP2-7





TP2-8

CHANGELOG.md

```
# Journal des changements

## Non publié

- message-b
- message-a

- Version initiale de l'API tasks-api.

```

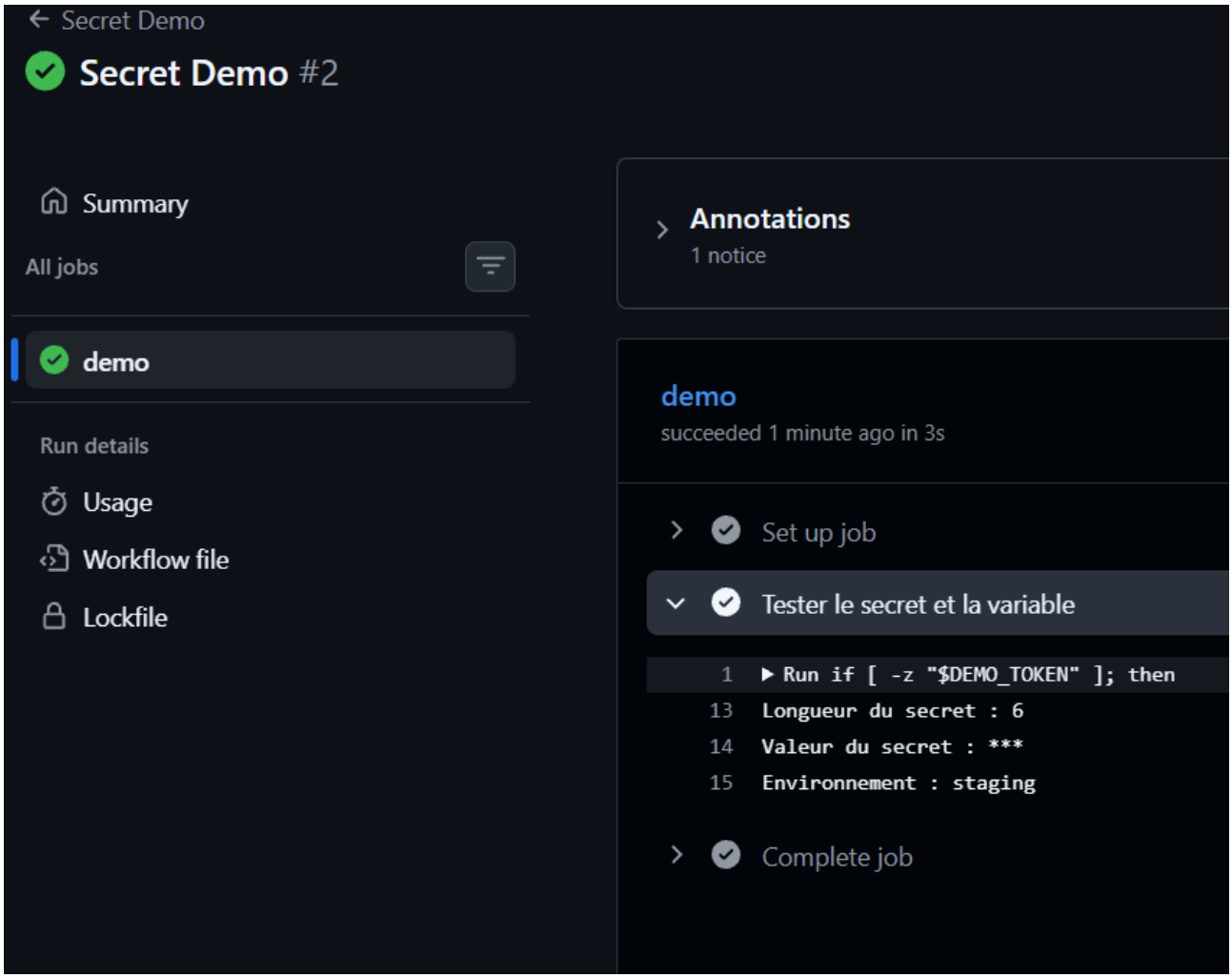
Pourquoi Git ne peut-il pas résoudre ce conflit seul ?

car il y à une modification de la même ligne, et git ne peux pas savoir la quelle garder.

Qu'est-ce qui a vérifié le code après la résolution ?

La CI avec les lint et les tests

TP2-9



Pourquoi GitHub masque-t-il le secret ?

pour éviter qu'ils soient exposés publiquement ou accessibles à des personnes non autorisées

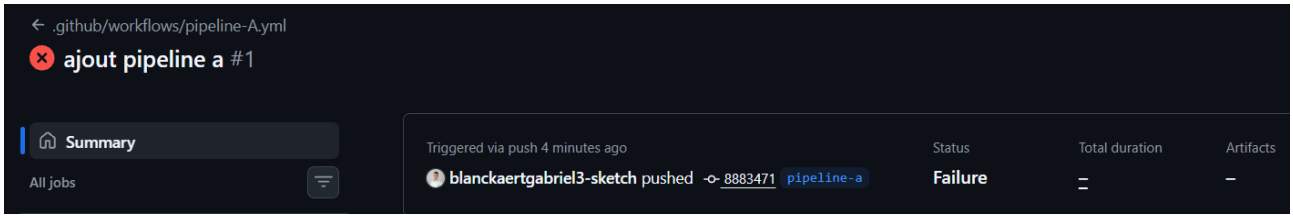
Que se passe-t-il si on le commite dans le dépôt ?

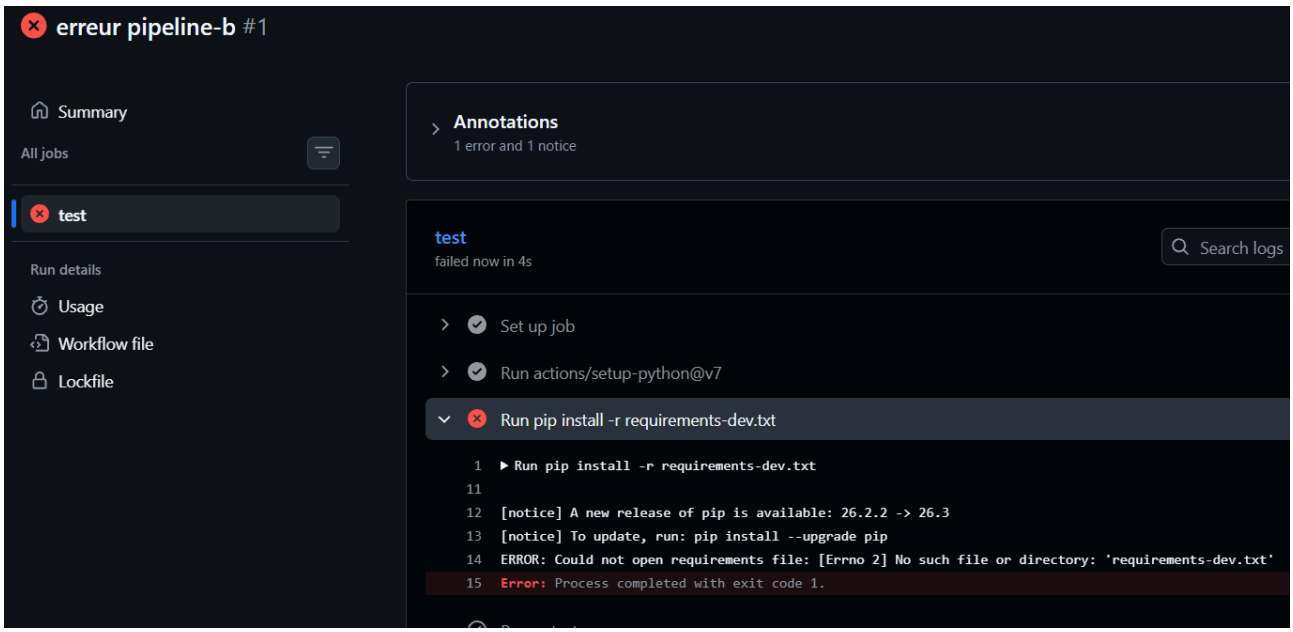
Le secret devient accessible aux personnes ayant accès au dépôt

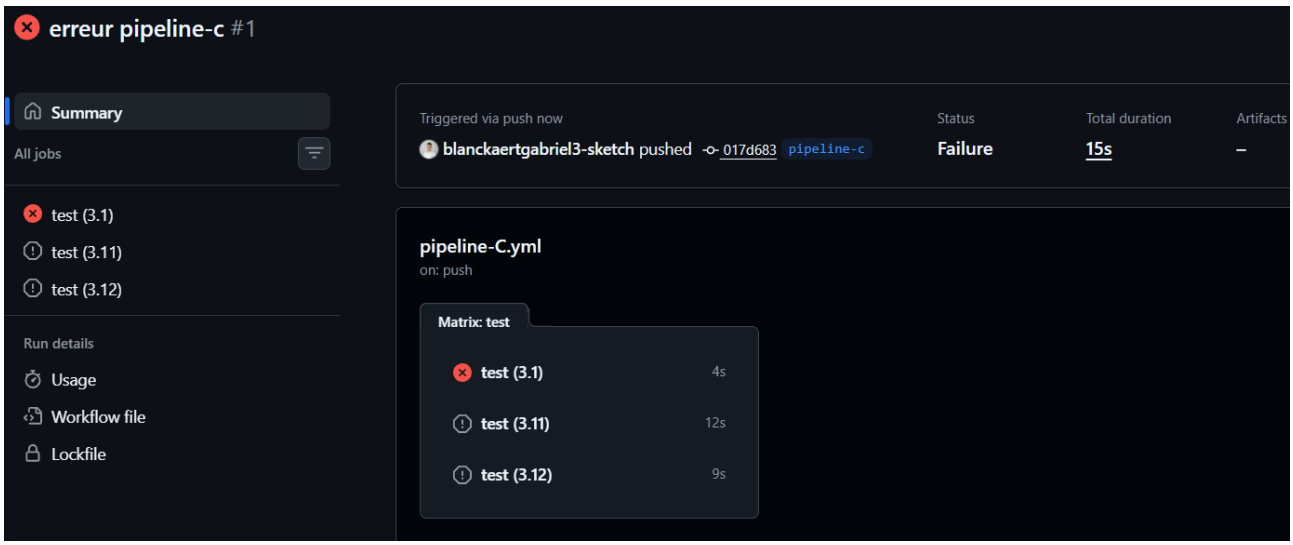
Secret ou variable : comment choisir ?

- Secret : pour les informations sensibles comme les mots de passe, les clés API et les jetons d'accès.
- Variable : pour les informations non sensibles comme le nom de l'environnement (`staging`), le mode de fonctionnement ou la configuration de l'application.

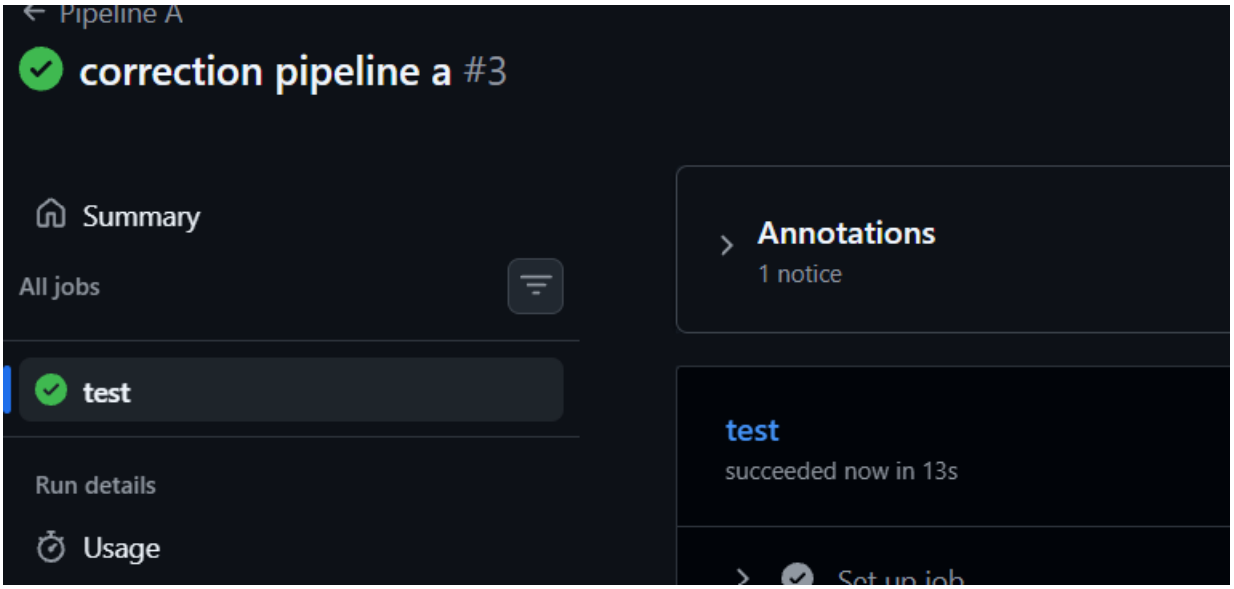
TP2-10



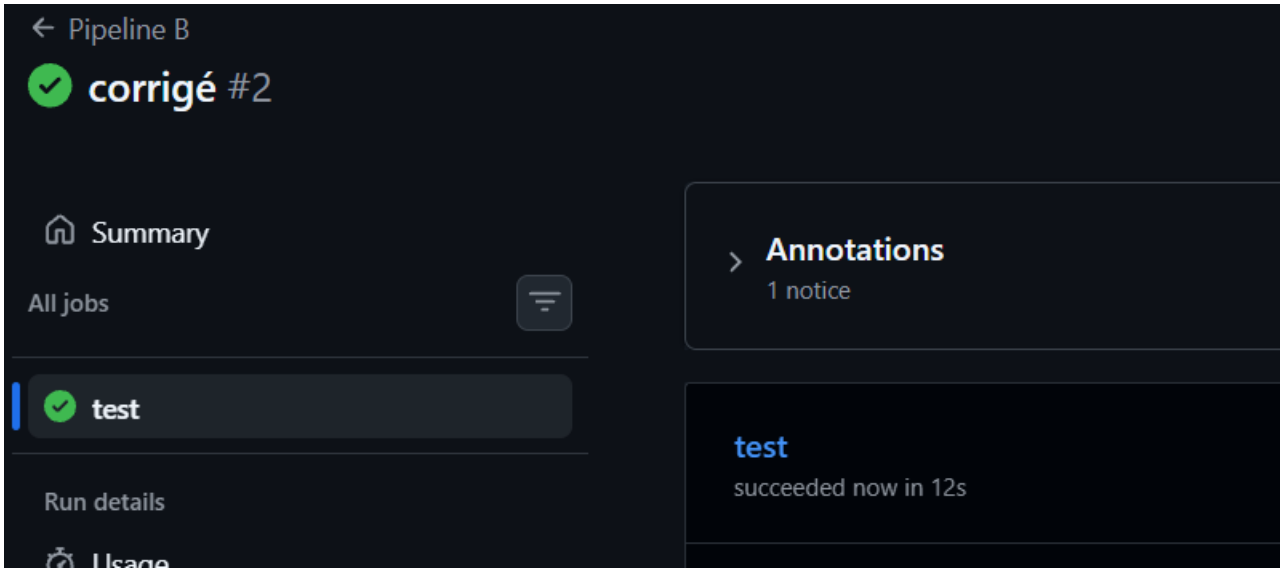

erreur pipeline-b #1
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 error and 1 notice
test
failed now in 4s
Search logs
Set up job
Run actions/setup-python@v7
Run pip install -r requirements-dev.txt
1 ▶ Run pip install -r requirements-dev.txt
11
12 [notice] A new release of pip is available: 26.2.2 -> 26.3
13 [notice] To update, run: pip install --upgrade pip
14 ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements-dev.txt'
15 Error: Process completed with exit code 1.


erreur pipeline-c #1
Summary
All jobs
test (3.1)
test (3.11)
test (3.12)
Run details
Usage
Workflow file
Lockfile
Triggered via push now
blanckaertgabriel3-sketch pushed 017d683 pipeline-c
Status
Failure
Total duration
15s
Artifacts
–
pipeline-C.yml
on: push
Matrix: test
test (3.1) 4s
test (3.11) 12s
test (3.12) 9s

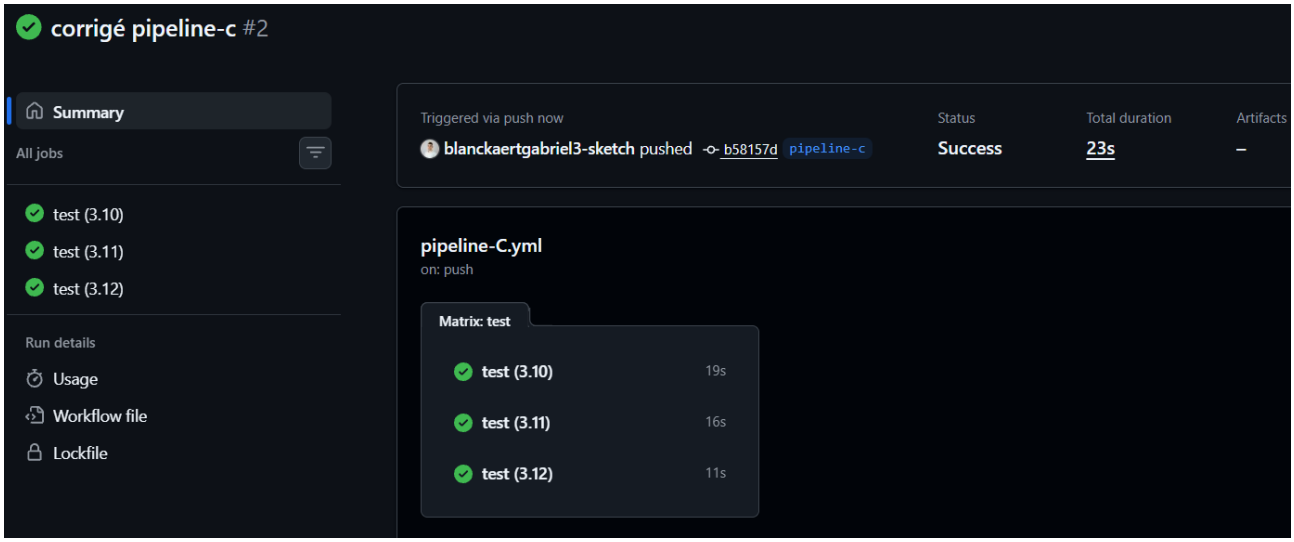
TP2-11



l'erreur de pipeline-A était l'indentation pour le run ligne 12



il manquais une ligne
- uses: actions/checkout@v6



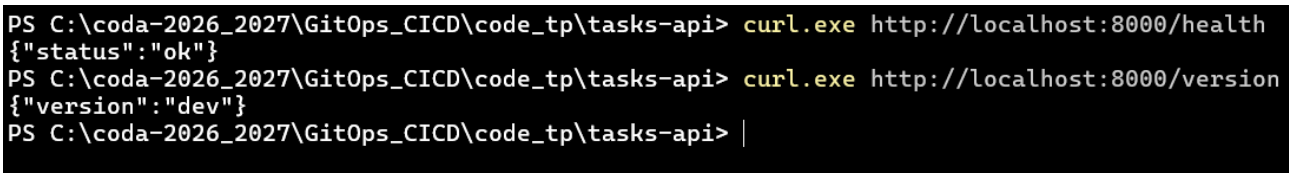
il manquais les guillemets pour la version de python avec matrix

```
python-version: ["3.10", "3.11", "3.12"]
```

TP3

TP3-1

```
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$ docker build -t tasks-api:local .
[+] Building 27.0s (11/11) FINISHED                                                
```



docker images tasks-api : notez la taille de l'image ;

la taille de l'image est : **201MB**

modifiez un fichier de app/, reconstruisez : quelles étapes sont « CACHED »

`FROM`

- `WORKDIR`
- `COPY requirements.txt .`
- `RUN pip install ...`

Pourquoi requirements.txt est-il copié avant le code ?

pour utiliser le cache de docker

Différence entre une image et un conteneur ?

une image est le modèle contenant l'application, le conteneur permet d'exécuter l'application de manière isolé.

À quoi sert le .dockerignore ?

permet d'exclure certains fichiers lors de la création du conteneur.

TP3-2

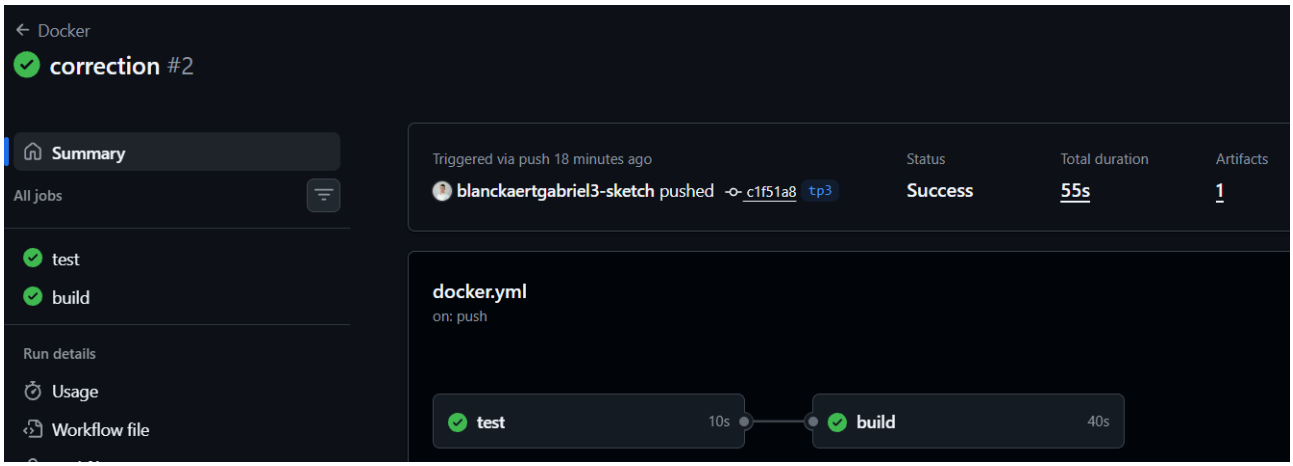
```
PS C:\coda-2026_2027\GitOps_CICD\code_tp\tasks-api> docker compose up --build
[+] Building 2.6s (13/13) FINISHED
 => [internal] load local bake definitions                                                                                0.1s
 => => reading from stdin 548B                                                                                            0.1s
 => [internal] load build definition from Dockerfile                                                                      0.1s
 => => transferring dockerfile: 1.38kB                                                                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                       1.0s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 196B                                                                                         0.0s
 => [1/6] FROM docker.io/library/python:3.12-slim@sha256:b18b261e1b4974b4f321451ad5251ded85abb22c2b8b262da527dda1d031ab30  0.1s
 => => resolve docker.io/library/python:3.12-slim@sha256:b18b261e1b4974b4f321451ad5251ded85abb22c2b8b262da527dda1d031ab30  0.1s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 363B                                                                                         0.0s
 => CACHED [2/6] WORKDIR /app                                                                                             0.0s
 => CACHED [3/6] COPY requirements.txt .                                                                                  0.0s
 => CACHED [4/6] RUN pip install --no-cache-dir -r requirements.txt                                                       0.0s
 => CACHED [5/6] COPY app/ app/                                                                                           0.0s
 => CACHED [6/6] RUN useradd --create-home --uid 10001 appuser                                                            0.0s
 => exporting to image                                                                                                    0.3s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:845a021dcc42d99da32272c14dfe6f584a856a08ae9b8755433dd714fef28034                         0.0s
 => => exporting config sha256:93ba0a524fd7fd3ace9366c4f8df3f6d992ba991a473c2ff053b41a4996c3565                           0.0s
 => => exporting attestation manifest sha256:dc3480bea5df9058364e1913c7d195a14c5db7535398afe693eeca50408b7f44             0.1s
 => => exporting manifest list sha256:24cafd6e73988b5fcfd24a8aa68b4521fc47a7dad95abb5508815d112ff4d66f                    0.0s
 => => naming to docker.io/library/tasks-api-api:latest                                                                   0.0s
 => => unpacking to docker.io/library/tasks-api-api:latest                                                                0.0s
 => resolving provenance for metadata file                                                                                0.0s
[+] Running 2/2
 ✔tasks-api-api              Built                                                                                        0.0s
 ✔Container tasks-api-api-1  Recreated                                                                                    0.4s
Attaching to api-1
api-1  | [2026-11-04 12:40:49 +0000] [1] [INFO] Starting gunicorn 26.2.0
api-1  | [2026-11-04 12:40:49 +0000] [1] [INFO] Listening at: http://0.0.0.0:8000 (1)
api-1  | [2026-11-04 12:40:49 +0000] [1] [INFO] Using worker: sync
api-1  | [2026-11-04 12:40:49 +0000] [7] [INFO] Booting worker with pid: 7
api-1  | [2026-11-04 12:40:49 +0000] [1] [INFO] Control socket listening at /home/appuser/.gunicorn/gunicorn.ctl
```

```
PS C:\coda-2026_2027\GitOps_CICD\code_tp\tasks-api> curl.exe http://localhost:8000/version
{"version":"local"}
```

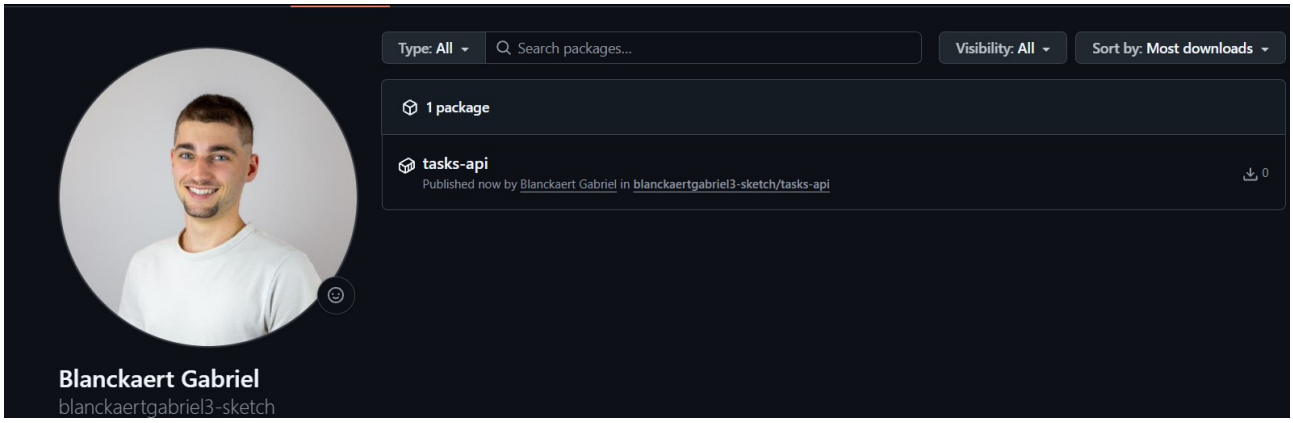
Que gagne-t-on à décrire le lancement dans un fichier plutôt que dans une longue commande ?

toute l'équipe peut lancer l'app simplement, en utilisant la commande : docker compose up.

TP3-3



TP3-4



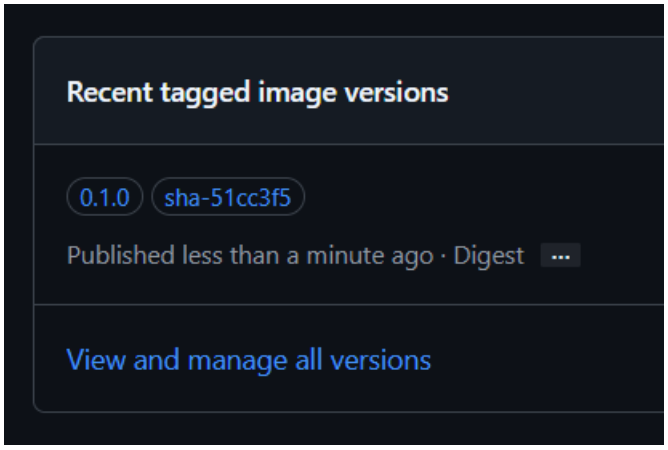
Pourquoi `build` attend-il `test` ?

Pour éviter de construire et publier une image Docker si les tests échouent, ce qui garantit que seules les versions validées sont déployées.

Pourquoi éviter le tag `latest` en production ?

Parce qu'il ne permet pas d'identifier précisément une version et peut changer à chaque publication.

TP3-5



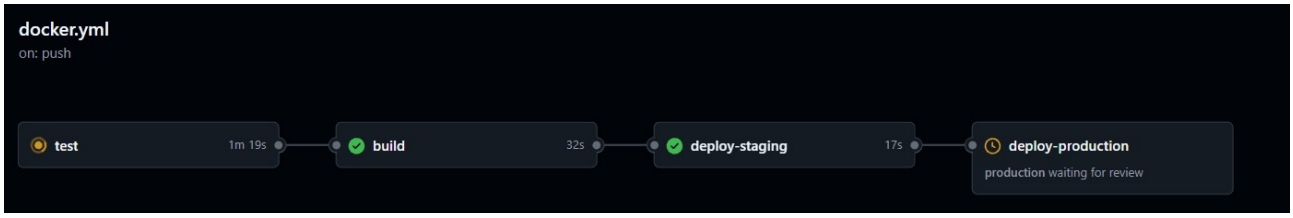
TP3-6

```
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$ docker run --rm -p 8000:8000 ghcr.io/blanckaertgabriel3-sketch/tasks-api:0.1.0
[2026-11-04 14:21:03 +0000] [1] [INFO] Starting gunicorn 26.2.0
[2026-11-04 14:21:03 +0000] [1] [INFO] Listening at: http://0.0.0.0:8000 (1)
[2026-11-04 14:21:03 +0000] [1] [INFO] Using worker: sync
[2026-11-04 14:21:03 +0000] [8] [INFO] Booting worker with pid: 8
[2026-11-04 14:21:03 +0000] [1] [INFO] Control socket listening at /home/appuser/.gunicorn/gunicorn.ctl
```

Quel est l'avantage d'une image versionnée par rapport à « le code de ma machine » ?

Elle garantit de déployer exactement la même version de l'application sur toutes les machines.

TP3-7





Test de fumée : vérifier rapidement que l'application fonctionne après son déploiement.

Approbation humaine : éviter qu'une version non validée soit déployée en production.

TP3-8

```
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$ docker run --rm tasks-api:local id
uid=10001(appuser) gid=10001(appuser) groups=10001(appuser)
```

Pourquoi ne pas exécuter une application en root dans un conteneur ?

Pour limiter les dégâts en cas de compromission du conteneur.

TP4

TP4-1

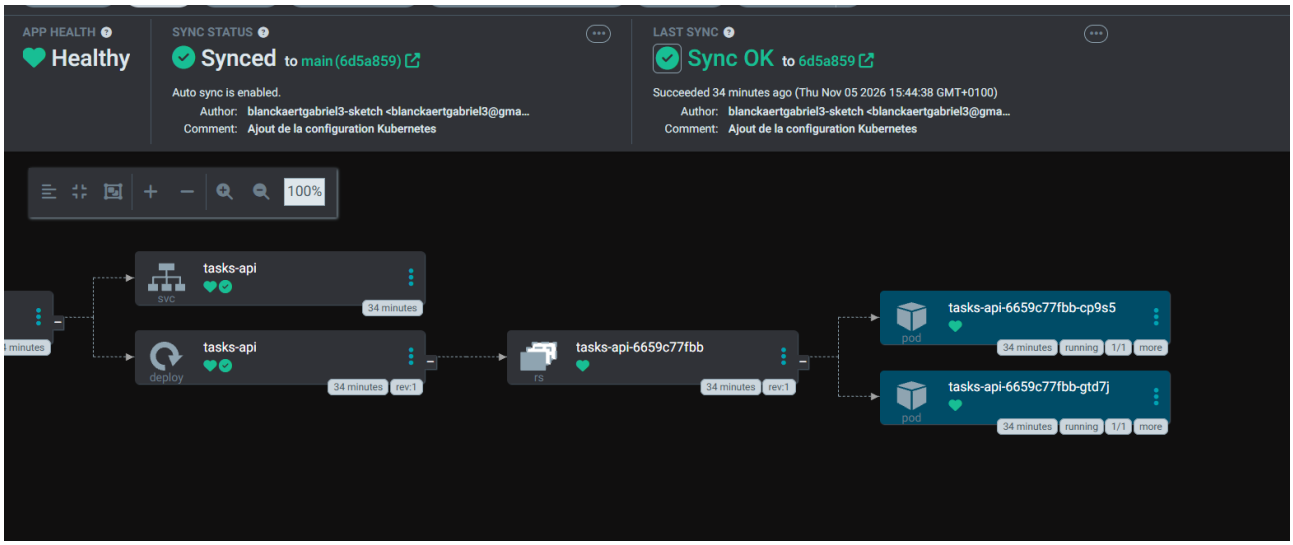
```
PS C:\coda-2026_2027\GitOps_CICD\code_tp\tasks-api> kubectl get nodes
NAME                   STATUS   ROLES           AGE   VERSION
gitops-control-plane   Ready    control-plane   22h   v1.34.0
```

TP4-2

```
PS C:\coda-2026_2027\GitOps_CICD\code_tp\tasks-api> kubectl get pods -n argocd
NAME                                                READY   STATUS    RESTARTS   AGE
argocd-application-controller-0                     1/1     Running   0          77s
argocd-applicationset-controller-c89dbf54b-zbblq    1/1     Running   0          78s
argocd-dex-server-79c78d477-vmbkc                   1/1     Running   0          78s
argocd-notifications-controller-9d797fbb4-hcp5m     1/1     Running   0          78s
argocd-redis-69cfc578c9-nfbff                       1/1     Running   0          78s
argocd-repo-server-8647f8f6fb-ddhrk                 1/1     Running   0          78s
argocd-server-9bff5b968-fd8rc                       1/1     Running   0          78s
PS C:\coda-2026_2027\GitOps_CICD\code_tp\tasks-api>
```

TP4-3

```
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$ kubectl get pods -n tasks -w
NAME                         READY   STATUS              RESTARTS   AGE
tasks-api-6659c77fbb-cp9s5   0/1     Pending             0          0s
tasks-api-6659c77fbb-gtd7j   0/1     Pending             0          0s
tasks-api-6659c77fbb-gtd7j   0/1     Pending             0          0s
tasks-api-6659c77fbb-cp9s5   0/1     Pending             0          0s
tasks-api-6659c77fbb-gtd7j   0/1     ContainerCreating   0          0s
tasks-api-6659c77fbb-cp9s5   0/1     ContainerCreating   0          1s
tasks-api-6659c77fbb-gtd7j   0/1     Running             0          16s
tasks-api-6659c77fbb-cp9s5   0/1     Running             0          20s
tasks-api-6659c77fbb-gtd7j   1/1     Running             0          20s
tasks-api-6659c77fbb-cp9s5   1/1     Running             0          20s
```



TP4-4

```
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$ curl http://localhost:8082/version
{"version":"0.1.0"}
blanckaertgabriel@El-VICTOR:/mnt/c/coda-2026_2027/GitOps_CICD/code_tp/tasks-api$
```

TP4-5

TP4-6

TP4-7

TP4-8

TP4-9

TP4-10